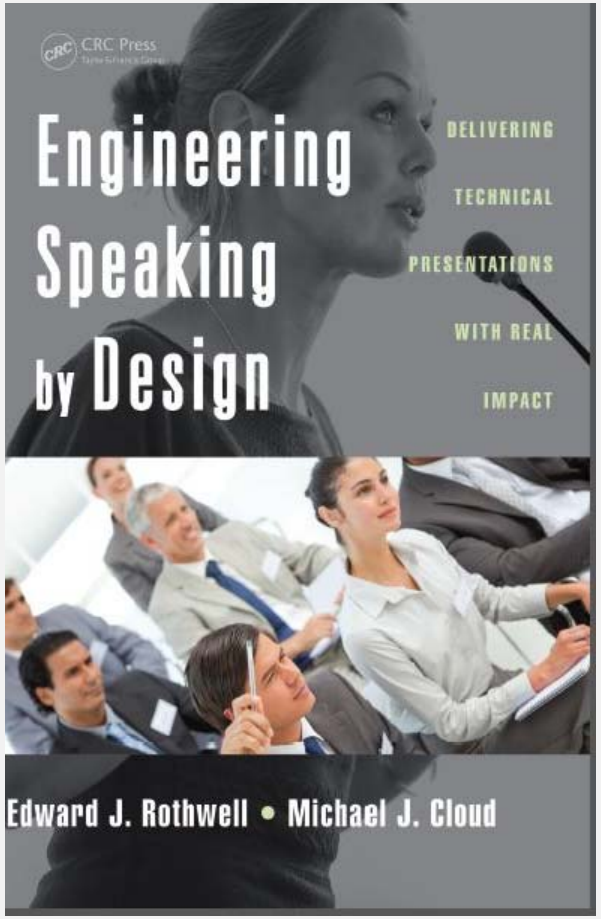
Fall 2019 VE496
Lecture notes
Week 7

Week 7 & 8 reading



The book ***Engineering Speaking***

concentrates only on “formal” engineering presentations:

- in-class presentations
- thesis or dissertation defenses
- presentations at professional conferences
- presentations to potential customers or funding agencies.

Presenting ≠ writing

- Audience-centric presentation
  - Understand-ability
  - Share-ability

Unique presentation challenges:

- Presenting highly technical findings and data

Audience-centric approach

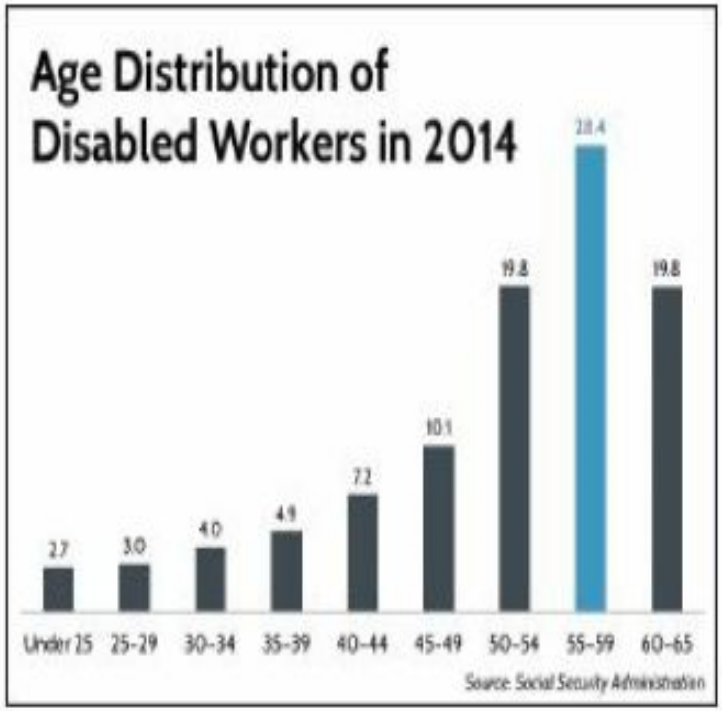
3 driving principles:

1. Visualize your content
2. Unify objects on your slides and throughout your presentation
3. Focus your audience's attention where you want it

1. Visualize your content

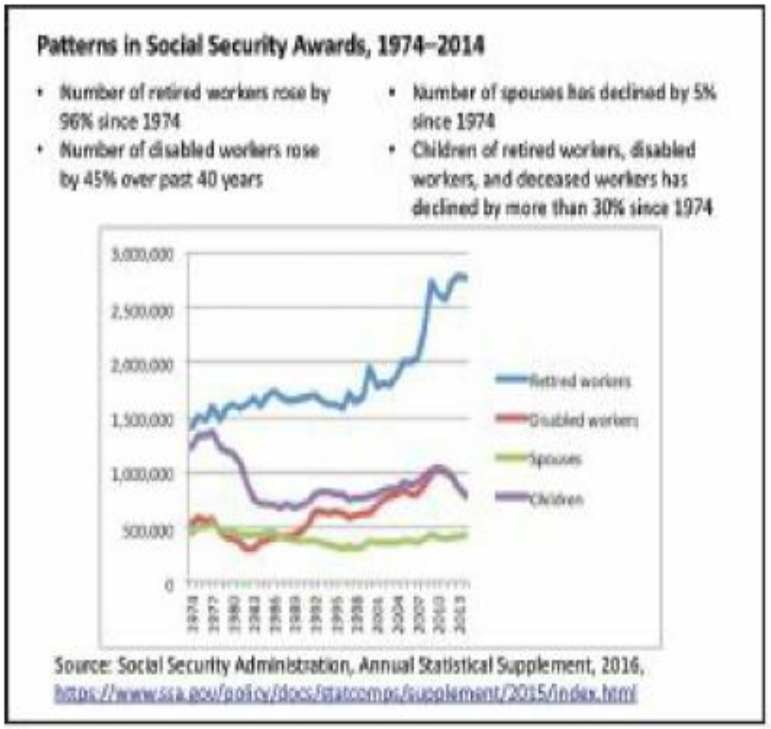
Number of Disabled Workers in 2014

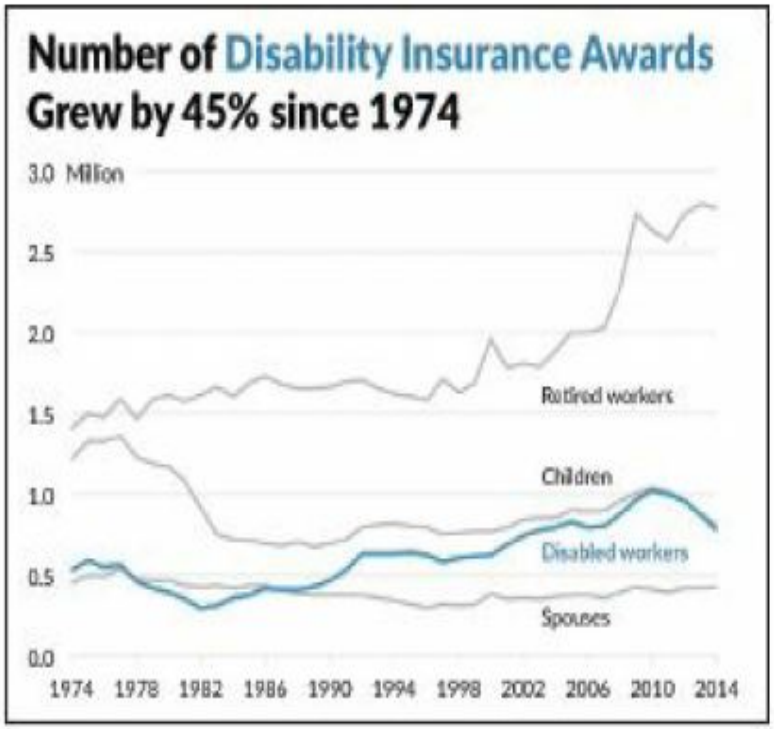
- There was less than 114,000 disabled workers under the age of 40 in 2014
- About 20% of total disabled workers were between 50 and 54 years old
- Another 28% were between 55 and 59 years old
- About 20% were between 60 and 65
  - Some of these workers moved from the disability program to the retirement program



2. Unify objects on your slides and throughout your presentation

3. Focus your audience's attention where you want it





Increasingly popular:

Presentation methods that involve drastically decreasing the amount of words and using abstract illustrations in their place

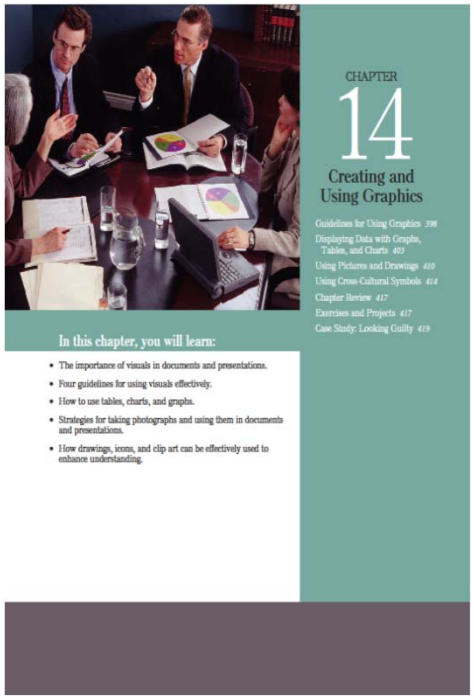
Visual literacy = Ability to create & interpret visual messages

Data
Information
Concepts
Ideas

Matching graphics to objectives

Communication Needs → Effective visual choices

Reference: Creating and using graphics



CHAPTER

14

Creating and Using Graphics

Guidelines for Using Graphics 398
Displaying Data with Graphs, Tables, and Charts 403
Using Pictures and Drawings 410
Using Cross-Cultural Symbols 414
Chapter Review 417
Exercises and Projects 417
Case Study: Looking Guilty 419

In this chapter, you will learn:

- The importance of visuals in documents and presentations.
- Four guidelines for using visuals effectively.
- How to use tables, charts, and graphs.
- Strategies for taking photographs and using them in documents and presentations.
- How drawings, icons, and clip art can be effectively used to enhance understanding.

The Best Data Visualization Tools of 2017

https://www.pcmag.com/roundup/346417/the-best-data-visualization-tools

The 38 best tools for data visualization

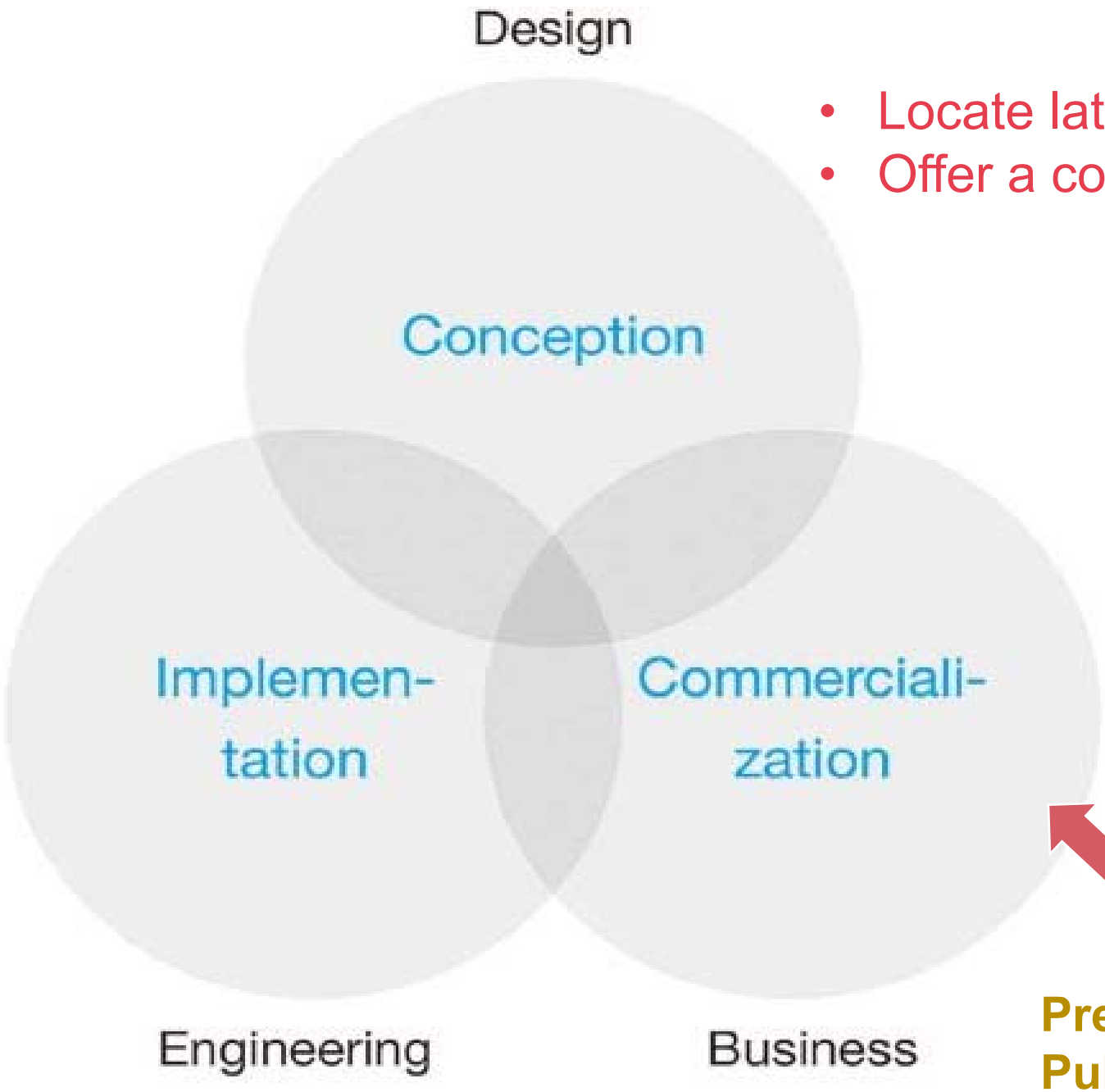
http://www.creativebloq.com/design-tools/data-visualization-712402

A periodic table of visualization methods

http://www.visual-literacy.org/periodic_table/periodic_table.html

10 Free Data Visualization Tools

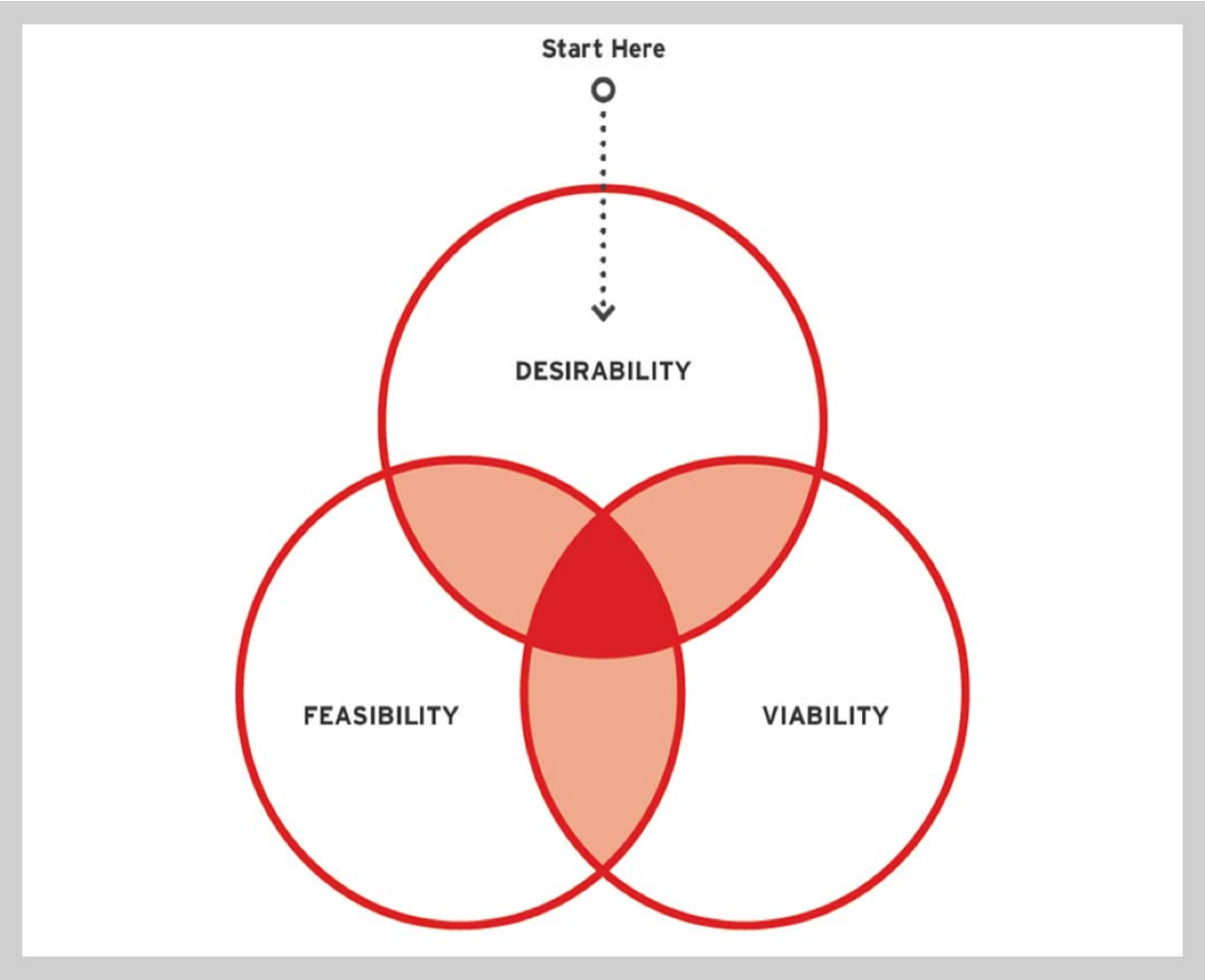
http://www.pcmag.com/article/347315/10-free-data-visualization-tools



- Locate latent needs
- Offer a concrete solution

Presentation
Public speaking

Hybrid professionals



✓ **Detailed specifications**

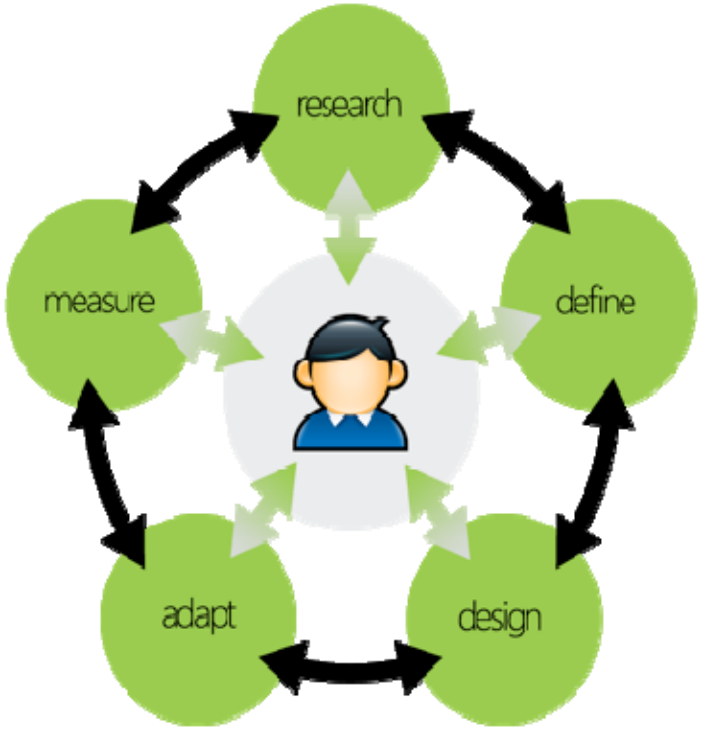
Functional overview

? **Emotional appeal**

Emotion = a powerful driver of consumer behavior

Me Me Me VS. You-focus

"*I'm here to save you from your helplessness and ignorance. If you only knew what I know, the world would be better place*."



Storytelling

Designers' presentations try to express a story as concretely and emotionally as possible.

You are not the hero
Your audience is the hero

The 4 elements making up the basic structure of a story:

1. The hero (the product user)
2. acquires a treasure (= product benefits)
3. by overcoming an obstacle (=everyday challenges)
4. and thereby becomes a little happier

Hybrid intellectual production

1. Empathy (user observation)
2. Symphony (synthesis)
3. Design (design in general)
4. Product story (storytelling)

Analysis

Empathy

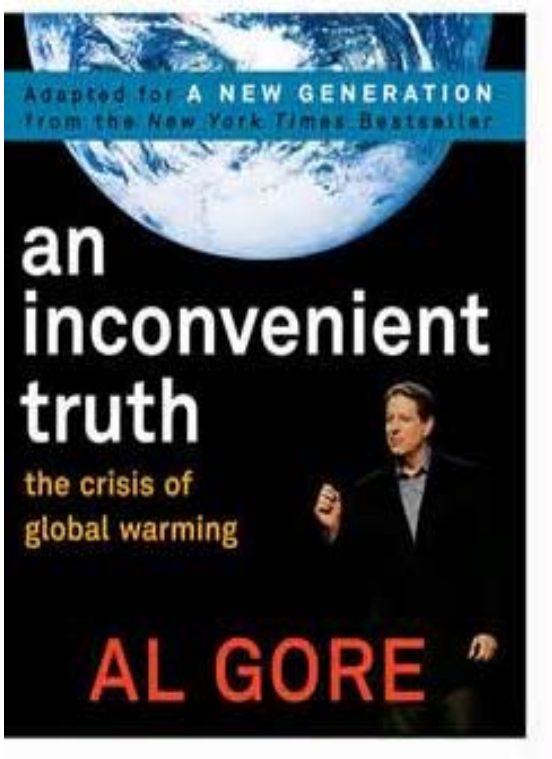
- User observation
- Design research
- Usability research

Synthesis

Metaphor & analogy

- Prototyping method
- Diagram design
- Communication design
- Storytelling

Analogy: Finding similarities between completely different things by making use of **metaphors**



"The planet has a fever"

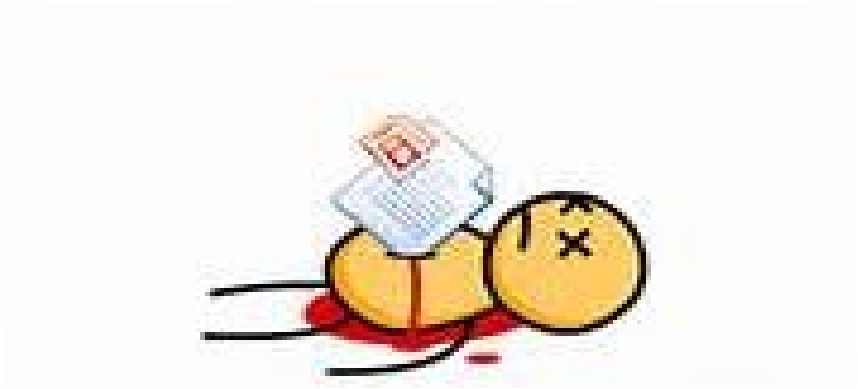
What is the metaphor of my new idea?

“1000 songs in your pocket”

Content
Visual impact
Delivery

“Death by PowerPoint”

- Lack of design awareness
- Schedule pressures
- Temptation to do things the same old way



David Philips offers 6 tips:

1. Deliver one message per slide
2. Avoid sentences in slides
3. Enlarge the size of the more important items
4. Use contrast in slides
5. Use dark background to shift attention to the presenter
6. Put no more than 6 objects in any one slide

Thank you!

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